

DBMS Quick Notes for B.Tech / Interview Preparation

1. Introduction to DBMS

A Database Management System (DBMS) is software used to create, manage, and manipulate databases. Advantages include reduced redundancy, data consistency, security, and backup/recovery.

2. ER Model

Entity, Attribute, Relationship. Types of attributes: Simple, Composite, Derived, Multivalued.

3. Keys

Super Key, Candidate Key, Primary Key, Alternate Key, Foreign Key, Composite Key.

4. Normalization

1NF: Atomic values. 2NF: Remove partial dependency. 3NF: Remove transitive dependency. BCNF: Every determinant should be a candidate key.

5. SQL Commands

DDL: CREATE, ALTER, DROP. DML: INSERT, UPDATE, DELETE. DQL: SELECT. TCL: COMMIT, ROLLBACK. DCL: GRANT, REVOKE.

6. Joins

INNER JOIN, LEFT JOIN, RIGHT JOIN, FULL OUTER JOIN, SELF JOIN.

7. Transactions

ACID Properties: Atomicity, Consistency, Isolation, Durability.

8. Concurrency Control

Locking, Two-Phase Locking (2PL), Deadlock, Starvation.

9. Indexing

Primary Index, Secondary Index, Clustered and Non-Clustered Index.

10. Important Interview Questions

Difference between DBMS and RDBMS, SQL vs NoSQL, Normalization, ACID Properties, Indexing, Joins, Triggers, Views, Stored Procedures.

Prepared for quick revision.